Problem Summary

- Given the roots of a binary tree, return the maximum depth. Maximum depth means the number of nodes along the longest path from the root down a leaf.

Key Idea

For each node:

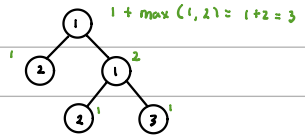
→ find the depth of the left Subtree

→ find the depth of the right Subtree

→ take the bigger one

→ add 1 for the current

Diagram Examples



Approach

→ DFS Recursion

1) If root is None, return 0

2) Recursively find left subtree depth

3) Recursively find right subtree depth

4) Return : 1+ max(left, right)

Time Complexity : $O(n)$ / $O(\log n)$ → Each node is visited once
(worst case / Balanced tree)

Space Complexity : $O(h)$ → Recursion stack, where h is tree height